

3.4 GPT and BERT

Same block, different mask and objective: next-token versus masked tokens.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

GPT and BERT are the same block with different **masks** and **objectives**. That choice is the difference between a generator and a bidirectional encoder.

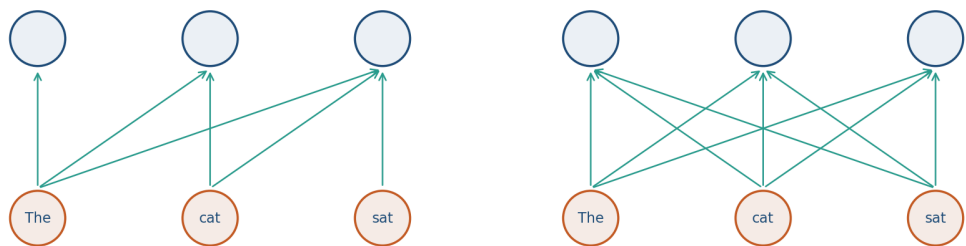
1. Two attention patterns

GPT (decoder-only): each token may attend to itself and the past. The pretraining job is **next-token** prediction. Sampling is ancestral: emit t , then condition on it.

BERT (encoder-only): every token may attend to every token. The pretraining job is **masked language modeling** (and originally next-sentence prediction). You cannot sample a paragraph left-to-right without extra machinery.

GPT: causal (next token)

BERT: bidirectional (masked token)



Causal mask: a lower-triangular pattern of legal scores. BERT’s “mask” is a **token** [MASK] in the input, not a future-key mask. Those two uses of the word confuse people all semester.

2. Objectives, in one table

	GPT	BERT
Mask	causal	none (random tokens hidden)
Loss	$p(x_t \mid x_{< t})$	$p(x_{\text{masked}} \mid x_{\text{rest}})$
Default use	generate, chat, complete	classify, tag, embed a span
Peek at the future?	no	yes, by design

Modern “BERT-like” encoders (RoBERTa, T5’s encoder, embedding models) keep the bidirectional idea. Modern LLMs are GPT-style stacks, often with extra alignment (Weeks 8–9).

Do not fine-tune BERT as if it were GPT. A [MASK] model is not a chat decoder. You *can* classify with a decoder via a prompt; you cannot stream tokens from BERT without a separate head.

3. Scaling, briefly

The same block, bigger: more layers, wider d , more heads, more data. Week 7 is scaling laws and efficiency. This week: if the mask is causal, you have a language model; if not, you have a contextual encoder. Do not fine-tune BERT as if it were GPT.

For the project: generation, RAG, and tool-use sit on a decoder. Classification into a small label set can still use an encoder, or a decoder with a prompt.

4. Teaching this note

30–40 minutes. Draw the two attention triangles, fill the objective table, then the 2×2 causal-softmax example. Play Karpathy **Let's build GPT 0:00–12:00** (tokens in, next token out). Then **0:00–8:00** of CodeEmporium BERT (MLM / [MASK]). If you only have one clip, keep Karpathy and assign BERT.

5. Worked example

Two tokens, raw scores $S = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

GPT (causal): illegal future is S_{12} . Set it to $-\infty$:

$$S_{\text{GPT}} = \begin{bmatrix} 1 & -\infty \\ 3 & 4 \end{bmatrix}.$$

Softmax row 1: $[1, 0]$. Row 2: $\text{softmax}([3, 4]) \approx [0.27, 0.73]$ because $e^3 \approx 20.1$, $e^4 \approx 54.6$.

BERT (bidirectional): softmax both rows of the original S . Row 1: $\text{softmax}([1, 2]) \approx [0.27, 0.73]$. Token 1 **does** look at token 2.

Objective contrast: GPT loss on this pair is $-\log p(x_2 | x_1)$. BERT, if token 1 is [MASK], is $-\log p(x_1 | x_2)$ (and any other masked sites).

6. Where students get stuck

- Using BERT when the demo must **stream** tokens.
 - Masking GPT scores with 0 instead of $-\infty$.
 - Calling every transformer “a GPT.”
-

7. Video

Watch [Karpathy: Let's build GPT](#) through the first attention / causal-mask implementation (start **0:00–12:00** in class; the attention code is later if you assign homework).

Also [CodeEmporium: BERT Neural Network — EXPLAINED!](#). Pause on masked language modeling and next-sentence prediction. You do not need to train BERT today.

8. Practice

1. Why is BERT the wrong pretrained model if your demo must stream tokens?
2. A sentiment classifier: encoder or decoder? Give one reason for each.
3. What does a causal mask set the illegal scores to before the softmax, and why not zero?
4. Scores $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, causal mask. Write A . (Uniform among **legal** keys.)
5. BERT hides 15% of tokens. In a sequence of length 20, how many tokens are masked in expectation?
If the loss is only on those positions, how many terms enter the mean that step?